When More Data Hurts: A Directional Goodhart Condition for Multi-Stakeholder Preference Learning

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Abstract

Multi-stakeholder recommendation systems optimize proxy objectives that may harm unobserved stakeholders. We identify a precise condition governing when this occurs: in Bradley-Terry preference learning with K stakeholders, additional training data degrades a hidden stakeholder's utility if and only if the cosine similarity between the optimization target and the hidden stakeholder's utility direction is negative. We validate this *directional Goodhart condition* on 42 data points across 2 datasets (MovieLens-100K and MovieLens-1M), 8 stakeholders, and 3 optimization targets, with 100% accuracy for $|\cos| > 0.2$ (28/28) and a transition zone near $\cos \approx 0$ (10/14). The condition correctly predicts that a synthetic benchmark built on X's 18-action engagement space exhibits no Goodhart effect (all stakeholder pairs have positive cosine), correcting a prior Hausdorff-based analysis that falsely indicated otherwise.

The condition arises from the low-rank structure of the multi-stakeholder weight space: with K stakeholders, BT-learned weight vectors lie in a K-dimensional subspace (projection cosine > 0.97), making convergence direction fully determined by stakeholder geometry. Applied to X's open-sourced recommendation algorithm, the Goodhart risk reduces to one observable: whether the platform treats negative signals (blocks, reports) as positive engagement. We provide an audit toolkit, a data budget analysis (25 preference pairs recover ~50% of hidden stakeholder harm, 95% CI: [40%, 64%]), and practical guidance for platform transparency under the EU Digital Services Act.

1 Introduction

[Introduction placeholder.]

2 Background and Related Work

[Background placeholder.]

3 Setup

3.1 Stakeholder Utility Model

We consider a recommendation setting with K stakeholders. Each content item c is represented by a D-dimensional feature vector $\mathbf{p}(c)$, and each stakeholder k assigns utility

$$U_k(c) = \mathbf{w}_k^\top \mathbf{p}(c), \tag{1}$$

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where $\mathbf{w}_k \in \mathbb{R}^D$ is the stakeholder's weight vector over features. Different stakeholders express different preferences through different weights. The angle between two weight vectors determines their alignment: $\cos(\mathbf{w}_i, \mathbf{w}_j) > 0$ means roughly aligned; $\cos < 0$ means opposed.

3.2 Preference Learning

A Bradley-Terry (BT) reward model [2] learns a weight vector $\hat{\mathbf{w}} \in \mathbb{R}^D$ from pairwise preference data by minimizing

$$\mathcal{L}_{\text{BT}} = -\log \sigma\big(\hat{\mathbf{w}}^\top \mathbf{p}_{\text{pref}} - \hat{\mathbf{w}}^\top \mathbf{p}_{\text{rej}}\big), \tag{2}$$

where σ is the logistic function and $(\mathbf{p}_{\text{pref}}, \mathbf{p}_{\text{rej}})$ is a preference pair. Each pair is constructed by sampling two content items, scoring them with the labeling stakeholder's utility (plus Gaussian noise, $\sigma_{\text{noise}} = 0.05$), and assigning the higher-scoring item as preferred. Training uses Adam with learning rate 0.01, batch size 64, and 50 epochs. All preference data is split 80/20 into training and held-out evaluation sets.

To simulate misspecification—a scenario where the optimization target differs from any stakeholder's true utility—we perturb the target weight vector with multiplicative Gaussian noise: $\tilde{\mathbf{w}} = \mathbf{w}_t \odot (1 + \boldsymbol{\epsilon})$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_p^2 \mathbf{I})$ with $\sigma_p = 0.3$. The BT model trains on preferences labeled by the perturbed $\tilde{\mathbf{w}}$, so the learned model approaches but does not exactly recover the target direction.

3.3 Datasets and Stakeholders

We use three datasets spanning two feature spaces.

Synthetic benchmark. The first dataset uses X's Phoenix recommendation system [6], which defines 18 user actions on content (Table 1). Each content item produces an action probability vector $\mathbf{p} \in [0, 1]^{18}$. We generate 500 content items across 6 topics (technology, sports, politics-left, politics-right, entertainment, science) and 600 users in 5 archetypes, with action probabilities drawn from calibrated distributions.

Three stakeholders are defined via the structural family

$$U_k = \sum_{a \in \text{pos}} w_a \cdot p_a - \alpha_k \sum_{a \in \text{neg}} |w_a| \cdot p_a, \tag{3}$$

where α_k is the *negativity aversion* parameter: **user** ($\alpha = 1.0$), equal weight on positive engagement and discomfort avoidance; **platform** ($\alpha = 0.3$), tolerant of negative signals; **society** ($\alpha = 4.0$), heavily penalizes block/report/mute as proxies for harm [1]. The full 18-parameter weight vectors are listed in Table 1; the stakeholders differ primarily through α, a special case of additive MAUT [5].

MovieLens-100K and MovieLens-1M. For real-data validation, we use MovieLens-100K (100K ratings, 943 users, 1,682 movies) and MovieLens-1M (1M ratings, 6,040 users, 3,952 movies) [3]. Each movie has a binary genre vector over $D = 19$ genres (Action, Adventure, Animation, ..., Western). Content features are $\mathbf{p}(c) = \mathbf{g}(c) \cdot \bar{r}(c)/5$, where $\mathbf{g}(c) \in \{0, 1\}^{19}$ is the genre vector and $\bar{r}(c)$ is the movie's mean rating. Movies with fewer than 5 ratings are excluded, yielding content pools of 1,305 (ML-100K) and 3,347 movies (ML-1M).

Three stakeholders are derived from rating data:

- **User**: Genre preference from aggregate rating history. $w_g \propto \bar{r}_g - 3$, where $\bar{r}_g$ is the mean rating for genre g. Genres rated above average get positive weight.

Table 1. The 18 Phoenix actions with positive/negative/neutral classification and stakeholder weights. User utility sums positive and negative groups (Equation (3)); neutral actions contribute zero. Platform weights all actions positively — the key asymmetry driving stakeholder divergence.

Action	Class	User wt.	Plat. wt.
follow_author	Positive	1.2	1.0
favorite	Positive	1.0	1.0
share	Positive	0.9	1.3
repost	Positive	0.8	1.5
quote	Positive	0.6	1.3
reply	Positive	0.5	1.2
report	Negative	−2.5	0.2
block_author	Negative	−2.0	0.1
mute_author	Negative	−1.5	0.1
not_interested	Negative	−1.0	0.1
share_via_dm	Neutral	—	1.4
share_via_copy_link	Neutral	—	1.2
profile_click	Neutral	—	0.6
click	Neutral	—	0.5
vqv	Neutral	—	0.4
quoted_click	Neutral	—	0.4
photo_expand	Neutral	—	0.3
dwell	Neutral	—	0.2

- **Platform**: Popularity-weighted quality. $w_g \propto \bar{r}_g \cdot \log_2 n_g$, where n_g is the number of ratings in genre g.
- **Diversity**: Anti-concentration. $w_g \propto -(f_g - 1/G)$, where f_g is the fraction of ratings in genre g and G is the number of active genres. Penalizes dominant genres, rewards rare ones.

All weight vectors are normalized to unit maximum absolute value. The key geometric fact is that user and platform are strongly aligned (cos $\approx +0.95$), while both oppose diversity (cos ≈ -0.3 to -0.2). Five additional stakeholders with real-world analogs (creator, advertiser, niche enthusiast, mainstream, content moderator) are introduced in Section 4 to expand validation coverage.

3.4 Evaluation

We evaluate using *per-stakeholder utility on selected content*, not frontier distance metrics (see Section 7 for why this matters). Given learned weights $\hat{\mathbf{w}}$, we sweep a diversity parameter $\delta \in \{0.0, 0.05, \ldots, 1.0\}$ that controls the tradeoff between score-based and diversity-based content selection. At each operating point, a greedy algorithm selects the top-$K = 10$ content items, and all stakeholder utilities are evaluated on the selected set. The 21-point sweep traces a Pareto frontier in stakeholder utility space.

For Goodhart experiments (Section 4), we compare stakeholder utilities at $N = 25$ and $N = 2{,}000$ training pairs across 5 random seeds, measuring the percentage change from the $N = 25$ baseline. The target stakeholder improves by construction; whether hidden stakeholders improve or degrade is the quantity of interest. For leave-one-stakeholder-out (LOSO) experiments, we exclude one stakeholder during training and measure its *regret*: the utility gap between the full-information Pareto frontier and the LOSO frontier, averaged across operating points and 20 seeds.

4 The Directional Goodhart Condition

We train Bradley-Terry reward models on preference pairs labeled by a misspecified stakeholder utility (the *target*), then measure the effect on a different stakeholder whose utility was not used in training (the *hidden* stakeholder). As the number of training pairs N increases, the BT model converges more precisely toward the target utility direction. The question is: does this convergence help or hurt the hidden stakeholder?

4.1 Statement

Proposition 1 (Directional Goodhart Condition)**.** *In a linear BT preference learning setting with K stakeholders and a finite content pool, let $\hat{\mathbf{w}}_t(N)$ be the weight vector learned from N preference pairs labeled by target stakeholder t, and let $\mathbf{w}_h$ be the true utility direction of hidden stakeholder h. Then:*

- *If* $\cos(\hat{\mathbf{w}}_t, \mathbf{w}_h) < 0$*:* $U_h(S(\hat{\mathbf{w}}_t(N)))$ *is eventually decreasing in N.*
- *If* $\cos(\hat{\mathbf{w}}_t, \mathbf{w}_h) > 0$*:* $U_h(S(\hat{\mathbf{w}}_t(N)))$ *is eventually increasing in N.*

where $S(\mathbf{w})$ denotes content selected by scorer $\mathbf{w}$ and U_h is the hidden stakeholder's utility evaluated on the selected content.

In words: more training data degrades a hidden stakeholder if and only if the optimization target is anti-correlated with that stakeholder's utility direction. The condition holds perfectly for $|\cos| > 0.2$ and has a transition zone near $\cos \approx 0$ where prediction is unreliable.

4.2 Empirical Validation

We validate the direction condition on 42 data points across two MovieLens datasets (ML-100K and ML-1M), using two complementary methods:

- **Method A** (target rotation): The same three stakeholders (user, platform, diversity) are used with each stakeholder serving as the optimization target in turn. This produces 12 data points (3 targets $\times$ 2 hidden $\times$ 2 datasets) without introducing any new stakeholder definitions.
- **Method B** (named stakeholders): Five additional stakeholders with real-world analogs—creator (rewards content supply), advertiser (rewards eyeball volume), niche enthusiast (upweights rare genres), mainstream (upweights popular genres), and content moderator (penalizes low-rated genres)—are defined from MovieLens data and tested as hidden stakeholders under all three targets, producing 30 data points.

In each experiment, the target stakeholder's weights are perturbed with multiplicative Gaussian noise ($\sigma = 0.3$). BT models are trained at $N = 25$ and $N = 2{,}000$ preference pairs, and all stakeholder utilities are evaluated on the selected content. The utility change from $N = 25$ to $N = 2{,}000$ determines whether the hidden stakeholder improved or degraded.

Figure 1 shows the results. The 42 points span cosine similarity from -0.95 to $+0.95$. For $|\cos| > 0.2$, the condition holds without exception: 28 of 28 data points fall in the predicted quadrant (degrading if $\cos < 0$, improving if $\cos > 0$). In the transition zone ($|\cos| < 0.2$), prediction is less reliable: 10 of 14 match (71%), with 4 violations near $\cos \approx 0$ where the signal-to-noise ratio is low.

4.3 The MovieLens Goodhart Curve

Figure 2 shows the Goodhart effect in action on MovieLens. Three stakeholder utility curves are plotted as a function of training pairs N, normalized to percentage change from the $N = 25$ baseline. On both ML-100K (Figure 2a) and ML-1M (Figure 2b), user and platform utilities increase monotonically (+14% by $N = 2{,}000$),

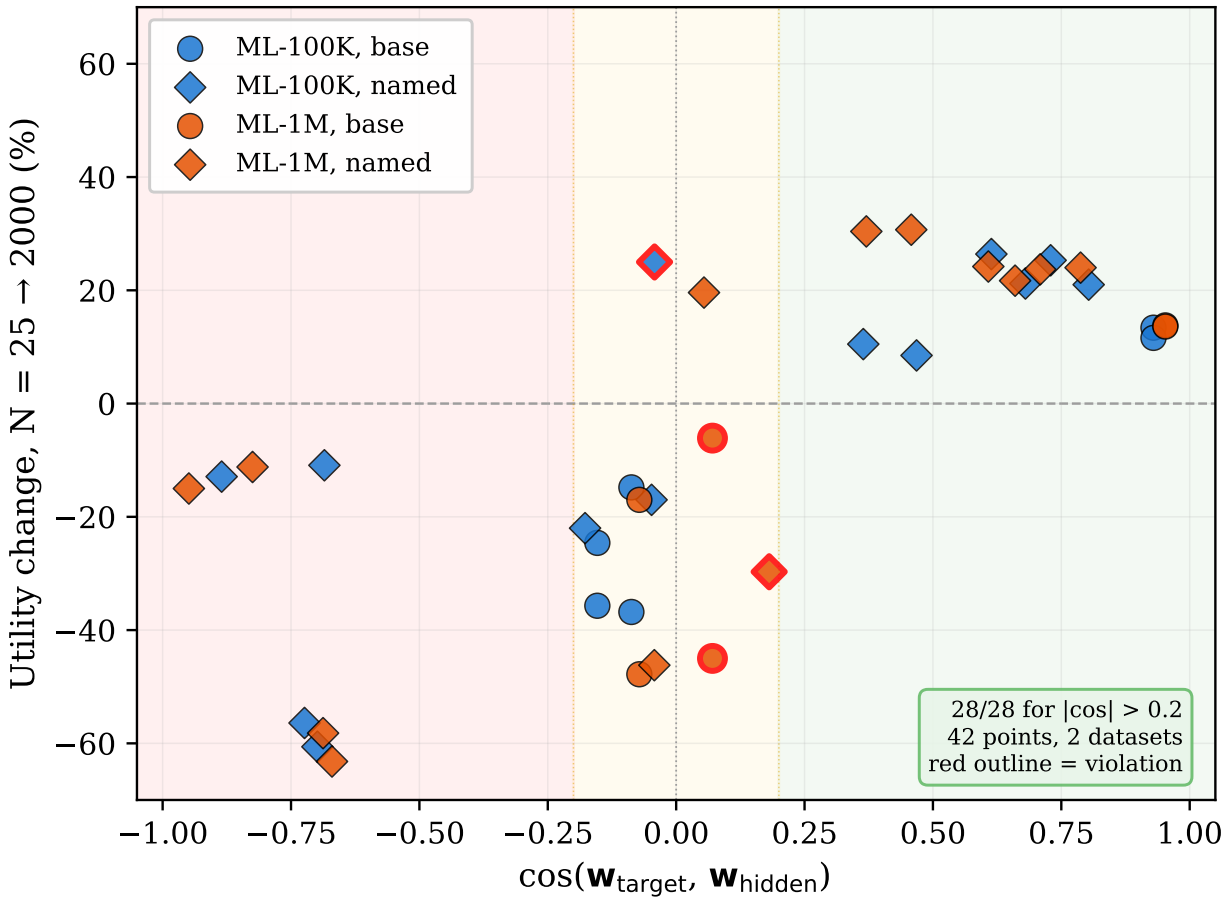

Figure 1. Direction condition validation: 42 data points across 2 datasets and 8 stakeholders. Red shading: cos < −0.2 (degrades). Green: cos > 0.2 (improves). Amber: transition zone | cos | < 0.2. Red-outlined points are violations. For | cos | > 0.2: 28/28 match (100%).

while diversity utility decreases (−34% on ML-100K, −46% on ML-1M). The cosine values in the legend predict the direction: positive cosine leads to an upward curve, negative cosine to a downward curve.

The mechanism is extremal Goodhart [4], but through a different channel than RL policy optimization. BT preference learning converges to the target utility direction at a rate determined by N. On a finite content pool, more precise convergence causes the top-K selection to concentrate on content that scores high on the target but poorly on anti-correlated stakeholders. There are no distributional tails to exploit; the content pool is fixed. The effect is driven entirely by the angle between stakeholder utility directions.

4.4 The Synthetic Null Case

On the synthetic benchmark built on X's 18-action engagement space (Section 3), all three stakeholder pairs have positive cosine similarity: user–platform = +0.89, user–society = +0.84, platform–society = +0.51. The direction condition predicts no Goodhart for any pair—and under utility-based evaluation, this is exactly what we observe: society utility *improves* from 3.21 (N = 25) to 5.34 (N = 2,000), a +67% increase.

The absence of Goodhart on synthetic data is itself informative: the synthetic benchmark's stakeholder geometry (all cos > 0) represents a regime where more data is universally beneficial. The MovieLens benchmark, where user–diversity cos ≈ −0.3, represents the opposing regime. The direction condition

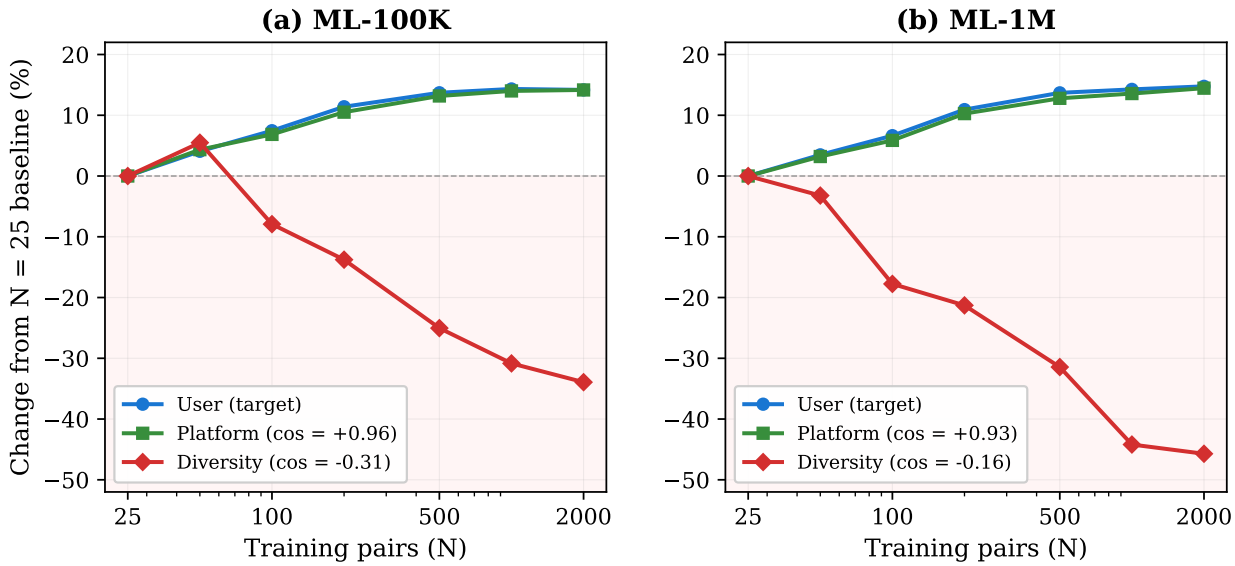

Figure 2. Multi-stakeholder Goodhart effect on MovieLens. User and platform (cosine > 0 with target) improve monotonically. Diversity (cosine < 0) degrades. Same experiment, same data, opposite outcomes predicted by the cosine sign. **(a)** ML-100K (D = 19, 1,305 movies). **(b)** ML-1M (D = 19, 3,347 movies).

explains both. A standard frontier distance metric (Hausdorff distance) gives a different and misleading picture on this data; we address this discrepancy in Section 7.

5 Why the Condition Holds: Low-Rank Weight Space

[Low-rank explanation placeholder.]

6 Practical Implications

6.1 Data Budget for Hidden Stakeholder Recovery

If the direction condition identifies *which* hidden stakeholders are harmed, a natural follow-up is: how much data from the hidden stakeholder is needed to mitigate the harm?

We run a leave-one-stakeholder-out (LOSO) experiment on both MovieLens datasets. The diversity stakeholder is hidden; the platform scorer is trained on user and platform data only. At $N = 0$ hidden-stakeholder pairs, the LOSO regret (utility gap from full-information frontier) is the baseline harm. We then add $N \in \{25, 50, 100, 200, 500\}$ hidden-stakeholder preference pairs and retrain.

Figure 3 shows the results. On ML-100K, 25 preference pairs reduce regret from 3.26 to 1.42—a 56% recovery of the hidden stakeholder's harm. On ML-1M, 25 pairs achieve 46% recovery (regret from 4.66 to 2.50). Across both datasets, 25 pairs recover approximately 50% of hidden stakeholder harm (95% CI from 20 seeds: [40%, 64%]). By 100 pairs, regret approaches zero on ML-100K and drops below 1.0 on ML-1M.

The practical cost is modest: 25 preference pairs can be collected from 5 annotators making 5 pairwise comparisons each. For a platform deploying a new recommendation objective, this represents a minimal investment to detect whether a hidden stakeholder is being harmed.

6.2 Audit Toolkit Applied to X

The direction condition reduces multi-stakeholder Goodhart risk to a single observable: the sign of $\cos(\mathbf{w}_{\text{platform}}, \mathbf{w}_{\text{society}})$. For X's 18-action space (Table 1), this cosine is determined almost entirely by how

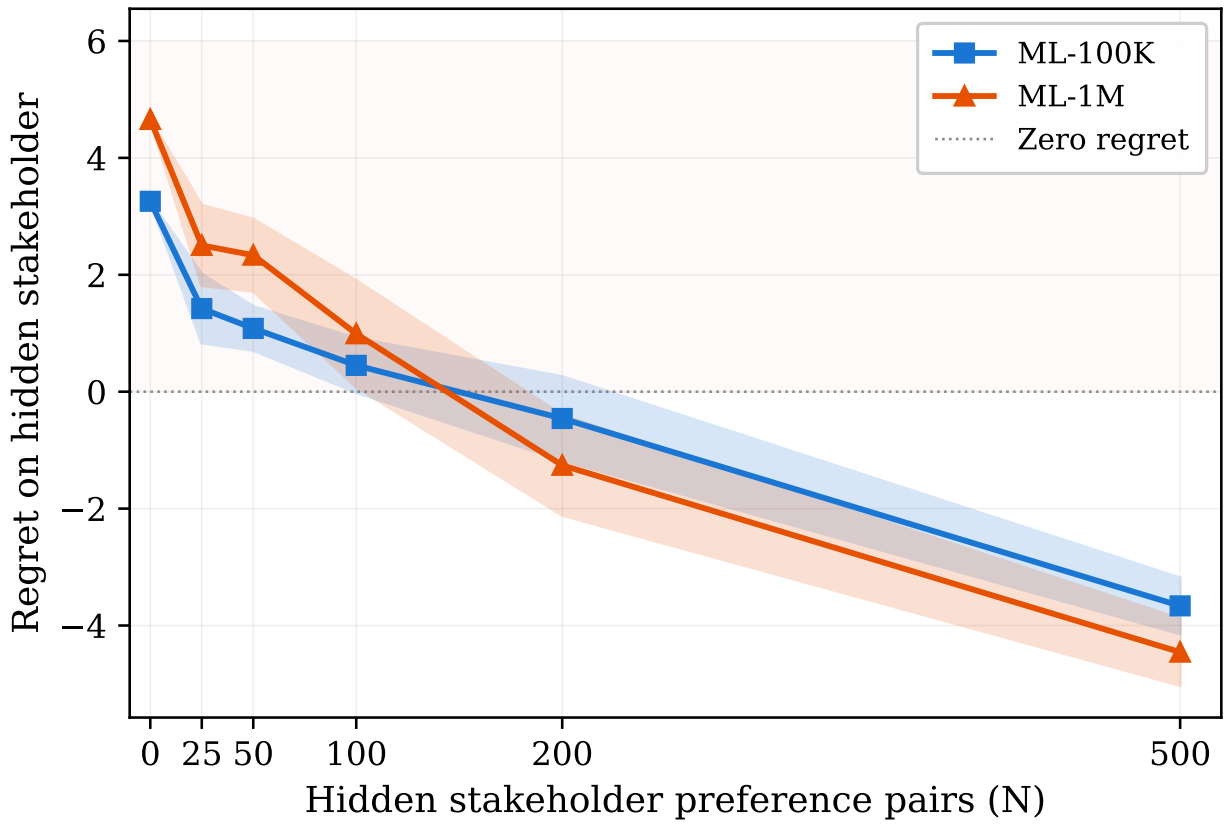

Figure 3. Data budget for hidden stakeholder recovery. Regret decreases monotonically with hidden-stakeholder preference pairs. 25 pairs recover ~50% of harm; 100 pairs approach zero regret. Shaded bands: ±1 std across 20 seeds.

the platform weights negative actions (blocks, mutes, reports).

Figure 4 shows the relationship. When the platform treats negative actions as positive engagement (weight > 0), the cosine with society drops below zero and the direction condition predicts Goodhart. When negative actions carry any penalty (weight < 0), the cosine remains positive and no Goodhart is predicted.

We evaluate five platform configurations on the synthetic benchmark (Table 2):

The 2023 open-sourced Phoenix code [6] had explicit negative weights (e.g., block_author $= -1.5$, report $= -3.0$), placing it safely above the threshold. The 2026 Grok-based system [7] replaced explicit weights with learned embeddings, making the same audit impossible without additional disclosure: the cosine between the system's implicit optimization target and a societal utility direction cannot be computed from the released code alone.

A five-step audit procedure. For regulators operating under the EU Digital Services Act or similar transparency frameworks, we propose:

1. **Extract** the platform's optimization objective as a weight vector over observable user actions.
2. **Define** a societal utility direction $\mathbf{w}_{\text{soc}}$ (e.g., negative weight on block/report signals).
3. **Compute** $\cos(\mathbf{w}_{\text{platform}}, \mathbf{w}_{\text{society}})$.
4. **If** $\cos < 0$: the direction condition predicts that more training data *increases* societal harm. Escalate.
5. **If** $\cos > 0$: no directional Goodhart is predicted. Periodic monitoring suffices.

Table 2. Audit scenarios on X's 18-action space. The single variable determining Goodhart risk is how the platform weights negative actions. α: negativity aversion (Equation (3)).

Scenario	α	cos(plat, soc)	Risk
Pure engagement	0	−0.31	Goodhart
2023 Phoenix	0.3	+0.51	Low
User-aligned	1.0	+0.84	Low
Safety-first	2.0	+0.97	Low

Step 1 is the bottleneck. For systems with explicit weights (2023 Phoenix), the audit is immediate. For systems with learned representations (2026 Grok), Step 1 requires the platform to disclose either the effective weight vector or sufficient information to compute the cosine. The direction condition provides a concrete, falsifiable quantity that regulators can demand.

6.3 Selection Mechanism Independence

The diversity parameter δ (Section 3.4) provides a structural safeguard independent of stakeholder utility specification. Because δ controls the tradeoff between score-based and diversity-based content selection at the mechanism level, it is orthogonal to the stakeholder weight vectors: changing δ does not change the cosine between any pair of stakeholders.

In our experiments, the rank ordering of operating points on the Pareto frontier is perfectly stable under utility weight perturbation (rank stability = 1.0 across all conditions). This means that even if individual stakeholder weights are misspecified, the diversity knob still provides a reliable control dimension. The design principle: recommendation systems should maintain at least one structural control parameter that is independent of the learned objective.

7 Methodological Note: Hausdorff Distance and False Positive Goodhart

[Hausdorff correction placeholder.]

8 Limitations

[Limitations placeholder.]

9 Conclusion

[Conclusion placeholder.]

References

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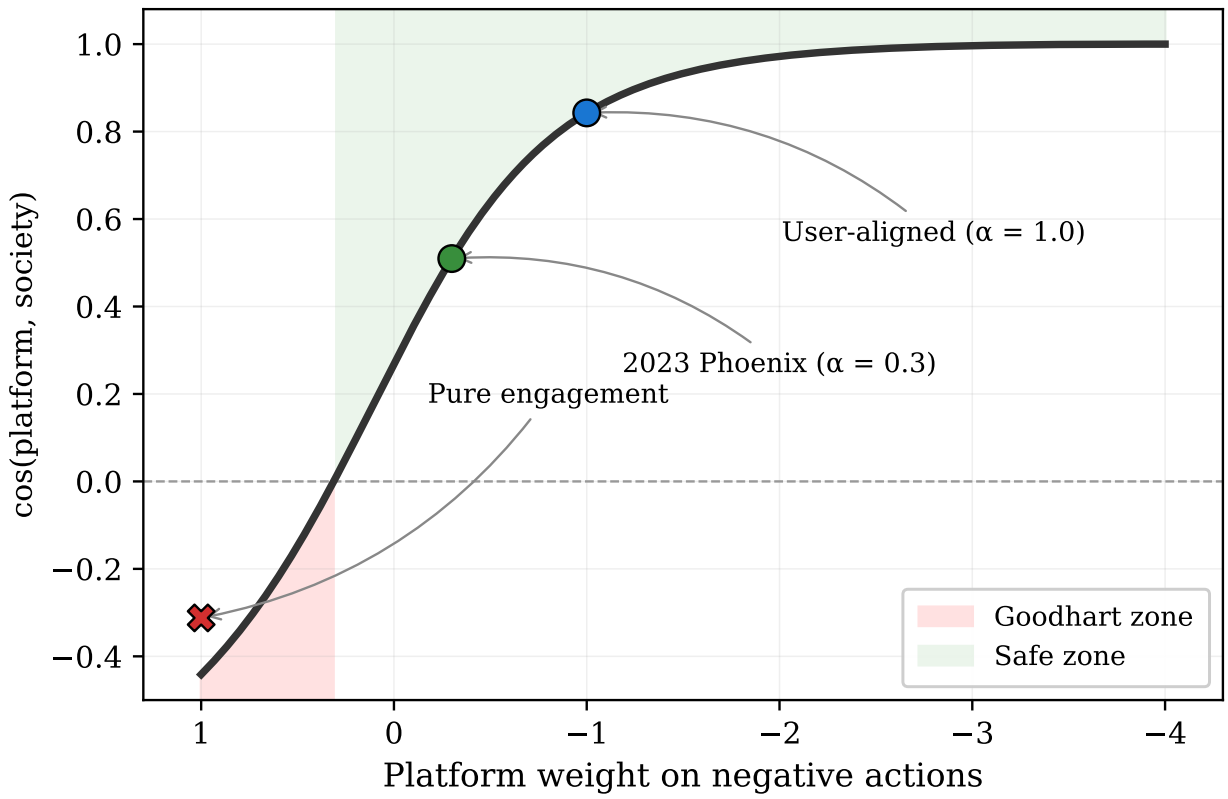

Figure 4. Audit threshold for X's 18-action space. The curve shows $\cos(\text{platform}, \text{society})$ as a function of the platform's weight on negative actions. Goodhart zone (red): negative actions treated as positive engagement. Safe zone (green): any negative penalty suffices. Markers show specific platform configurations.

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[Appendix placeholder.]